

Values of $\sin A^\circ$ and $\cos A^\circ$ in Radicals for Integral A

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Abstract

This paper develops a constructive radical framework for the exact evaluation of $\sin A^\circ$ and $\cos A^\circ$ for every integral angle A between 0° and 360° . The construction does not attack the high-degree minimal polynomial of $\cos 1^\circ$ directly. Instead, it exploits the algebraic structure of trigonometric addition, subtraction, double-angle, triple-angle, half-angle, complementary-angle, and quadrant identities.

The principal algebraic generators are obtained from the elementary angles 30° , 45° , 15° , and 18° , together with the cubic for $\cos 20^\circ$. Cardano's method produces the required cubic radicals, while subsequent half-angle and addition–subtraction operations generate 10° , 9° , 5° , 4° , 2° , and 1° . All remaining integral-degree values follow from finite algebraic operations and quadrant symmetries.

The resulting construction is interpreted as a finite algebraic generation problem, with connections to field theory, numerical verification, algorithms, data science, mathematical economics, finance, physics, chemistry, biology, psychology, political science, geopolitics, diplomacy, international relations, welfare economics, warfare economics, and systems engineering. These interdisciplinary interpretations are structural analogies rather than claims that trigonometric identities constitute empirical theories in those disciplines.

The paper ends with “The End”

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1 Introduction

The problem is to construct exact radical representations of

$$\sin A^\circ, \quad \cos A^\circ, \quad (1)$$

for every integral A in the interval

$$0 \leq A \leq 360. \quad (2)$$

A direct algebraic attack on $\cos 1^\circ$ leads to a polynomial of high degree. Such a direct approach obscures the much simpler compositional structure of the problem. The central idea of this paper is instead to construct a small set of algebraic generators and then use the trigonometric group law to generate the remaining values.

The method has four layers:

1. exact radical construction of a small set of anchor angles;
2. generation of smaller angles by half-angle formulae;
3. generation of intermediate angles by addition and subtraction;
4. extension from the first quadrant to the complete circle by complementary and quadrant symmetries.

The fundamental addition law is

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta, \quad (3)$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta. \quad (4)$$

The corresponding subtraction laws are

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta, \quad (5)$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta. \quad (6)$$

These identities constitute the principal computational engine of the construction.

2 Algebraic Foundations

2.1 The Pythagorean identity

For every real angle θ ,

$$\sin^2 \theta + \cos^2 \theta = 1. \quad (7)$$

Consequently,

$$\sin \theta = \pm \sqrt{1 - \cos^2 \theta}, \quad (8)$$

$$\cos \theta = \pm \sqrt{1 - \sin^2 \theta}. \quad (9)$$

For angles in the first quadrant, both signs are positive.

2.2 Double-angle identities

The double-angle identities are

$$\sin 2\theta = 2 \sin \theta \cos \theta, \quad (10)$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta, \quad (11)$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta, \quad (12)$$

$$\cos 2\theta = 2 \cos^2 \theta - 1. \quad (13)$$

Hence

$$\sin \theta = \sqrt{\frac{1 - \cos 2\theta}{2}}, \quad (14)$$

$$\cos \theta = \sqrt{\frac{1 + \cos 2\theta}{2}}, \quad (15)$$

for θ in the first quadrant.

2.3 Triple-angle identities

The triple-angle identities are

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta, \quad (16)$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta. \quad (17)$$

These identities transform certain trigonometric evaluation problems into cubic equations.

3 Elementary Radical Anchors

The familiar values are

$$\sin 30^\circ = \frac{1}{2}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}, \quad (18)$$

$$\sin 45^\circ = \frac{\sqrt{2}}{2}, \quad \cos 45^\circ = \frac{\sqrt{2}}{2}. \quad (19)$$

Using

$$15^\circ = 45^\circ - 30^\circ, \quad (20)$$

the subtraction formulae yield

$$\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}, \quad (21)$$

$$\cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}. \quad (22)$$

The pentagonal-angle identities yield

$$\cos 36^\circ = \frac{1 + \sqrt{5}}{4}, \quad (23)$$

$$\cos 18^\circ = \sqrt{\frac{1 + \cos 36^\circ}{2}} = \frac{\sqrt{10 + 2\sqrt{5}}}{4}. \quad (24)$$

Thus

$$\sin 18^\circ = \sqrt{\frac{1 - \cos^2 18^\circ}{1}} = \frac{\sqrt{5} - 1}{4}, \quad (25)$$

$$\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}. \quad (26)$$

4 Construction of $\cos 20^\circ$

Since

$$60^\circ = 3(20^\circ), \quad (27)$$

put

$$x = \cos 20^\circ. \quad (28)$$

The triple-angle identity gives

$$4x^3 - 3x = \cos 60^\circ = \frac{1}{2}. \quad (29)$$

Hence

$$8x^3 - 6x - 1 = 0. \quad (30)$$

Dividing by 8 produces the depressed cubic

$$x^3 - \frac{3}{4}x - \frac{1}{8} = 0. \quad (31)$$

For

$$x^3 + px + q = 0, \quad (32)$$

Cardano's discriminant is

$$\Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3. \quad (33)$$

Here,

$$p = -\frac{3}{4}, \quad (34)$$

$$q = -\frac{1}{8}, \quad (35)$$

and therefore

$$\Delta = \frac{1}{256} - \frac{1}{64} = -\frac{3}{256}. \quad (36)$$

Cardano's expression is consequently

$$\cos 20^\circ = \sqrt[3]{\frac{1+i\sqrt{3}}{16}} + \sqrt[3]{\frac{1-i\sqrt{3}}{16}}. \quad (37)$$

The two radicands are conjugates:

$$1+i\sqrt{3} = 2e^{i\pi/3}, \quad 1-i\sqrt{3} = 2e^{-i\pi/3}. \quad (38)$$

With conjugate cube-root branches,

$$\sqrt[3]{\frac{1+i\sqrt{3}}{16}} = \frac{1}{2}e^{i\pi/9}, \quad (39)$$

$$\sqrt[3]{\frac{1-i\sqrt{3}}{16}} = \frac{1}{2}e^{-i\pi/9}. \quad (40)$$

Their sum is real:

$$\cos 20^\circ = \frac{1}{2} \left(e^{i\pi/9} + e^{-i\pi/9} \right). \quad (41)$$

This is the classical casus irreducibilis phenomenon: a cubic with three real roots can require complex intermediate radicals in Cardano's formula.

Define the radical generator

$$R = \sqrt[3]{\frac{1+i\sqrt{3}}{16}} + \sqrt[3]{\frac{1-i\sqrt{3}}{16}}. \quad (42)$$

Then

$$R = \cos 20^\circ. \quad (43)$$

5 Construction of 10°

Since

$$20^\circ = 2(10^\circ), \quad (44)$$

the double-angle identity gives

$$R = 1 - 2\sin^2 10^\circ. \quad (45)$$

Putting

$$s_{10} = \sin 10^\circ, \quad (46)$$

we obtain the quadratic

$$2s_{10}^2 + R - 1 = 0. \quad (47)$$

The quadratic formula gives

$$s_{10} = \pm \sqrt{\frac{1-R}{2}}. \quad (48)$$

Since 10° lies in the first quadrant,

$$\boxed{\sin 10^\circ = \sqrt{\frac{1-R}{2}}}. \quad (49)$$

Similarly,

$$R = 2 \cos^2 10^\circ - 1, \quad (50)$$

so

$$\boxed{\cos 10^\circ = \sqrt{\frac{1+R}{2}}}. \quad (51)$$

Thus the cubic has been followed by a quadratic radical extension.

6 Construction of 9°

Since

$$18^\circ = 2(9^\circ), \quad (52)$$

the half-angle formulae give

$$\sin 9^\circ = \sqrt{\frac{1 - \cos 18^\circ}{2}}, \quad (53)$$

$$\cos 9^\circ = \sqrt{\frac{1 + \cos 18^\circ}{2}}. \quad (54)$$

Using the radical expression for $\cos 18^\circ$,

$$\sin 9^\circ = \sqrt{\frac{1}{2} - \frac{\sqrt{10+2\sqrt{5}}}{8}}, \quad (55)$$

$$\cos 9^\circ = \sqrt{\frac{1}{2} + \frac{\sqrt{10+2\sqrt{5}}}{8}}. \quad (56)$$

7 Construction of 5°

The logically non-circular construction of 5° is obtained directly from 10° :

$$5^\circ = \frac{10^\circ}{2}. \quad (57)$$

Consequently,

$$\sin 5^\circ = \sqrt{\frac{1 - \cos 10^\circ}{2}}, \quad (58)$$

$$\cos 5^\circ = \sqrt{\frac{1 + \cos 10^\circ}{2}}. \quad (59)$$

Substitution of the radical generator R gives

$$\sin 5^\circ = \sqrt{\frac{1}{2} - \frac{1}{2} \sqrt{\frac{1+R}{2}}}, \quad (60)$$

$$\cos 5^\circ = \sqrt{\frac{1}{2} + \frac{1}{2} \sqrt{\frac{1+R}{2}}}. \quad (61)$$

This establishes 5° without assuming prior knowledge of 1° .

8 Construction of 4°

Since

$$4^\circ = 9^\circ - 5^\circ, \quad (62)$$

the subtraction formulae give

$$\sin 4^\circ = \sin 9^\circ \cos 5^\circ - \cos 9^\circ \sin 5^\circ, \quad (63)$$

$$\cos 4^\circ = \cos 9^\circ \cos 5^\circ + \sin 9^\circ \sin 5^\circ. \quad (64)$$

Every term on the right-hand sides is already a radical expression.

9 Construction of 2°

Since

$$2^\circ = \frac{4^\circ}{2}, \quad (65)$$

we have

$$\sin 2^\circ = \sqrt{\frac{1 - \cos 4^\circ}{2}}, \quad (66)$$

$$\cos 2^\circ = \sqrt{\frac{1 + \cos 4^\circ}{2}}. \quad (67)$$

10 Construction of 1°

Finally,

$$1^\circ = \frac{2^\circ}{2}. \quad (68)$$

Therefore,

$$\sin 1^\circ = \sqrt{\frac{1 - \cos 2^\circ}{2}}, \quad (69)$$

$$\cos 1^\circ = \sqrt{\frac{1 + \cos 2^\circ}{2}}. \quad (70)$$

This gives a completely non-circular radical construction:

$$\boxed{20^\circ \longrightarrow 10^\circ \longrightarrow 5^\circ \longrightarrow 4^\circ \longrightarrow 2^\circ \longrightarrow 1^\circ.} \quad (71)$$

The 9° branch is independently obtained from 18° :

$$\boxed{18^\circ \longrightarrow 9^\circ.} \quad (72)$$

11 Independent Addition–Subtraction Construction of 3°

Since

$$3^\circ = 18^\circ - 15^\circ, \quad (73)$$

we have

$$\sin 3^\circ = \sin 18^\circ \cos 15^\circ - \cos 18^\circ \sin 15^\circ, \quad (74)$$

$$\cos 3^\circ = \cos 18^\circ \cos 15^\circ + \sin 18^\circ \sin 15^\circ. \quad (75)$$

Substitution gives

$$\sin 3^\circ = \frac{(\sqrt{5} - 1)(\sqrt{6} + \sqrt{2}) - \sqrt{10 + 2\sqrt{5}}(\sqrt{6} - \sqrt{2})}{16}, \quad (76)$$

$$\cos 3^\circ = \frac{\sqrt{10 + 2\sqrt{5}}(\sqrt{6} + \sqrt{2}) + (\sqrt{5} - 1)(\sqrt{6} - \sqrt{2})}{16}. \quad (77)$$

12 The Fundamental Radical Anchor Set

It is useful to define

$$S_A = \sin A^\circ, \quad C_A = \cos A^\circ. \quad (78)$$

The fundamental anchor set may then be written as

$$\boxed{\mathcal{A} = \{0, 1, 2, 3, 4, 5, 9, 10, 15, 18, 20, 30, 45\}}. \quad (79)$$

The corresponding values are generated by:

$$S_0 = 0, \quad C_0 = 1, \quad (80)$$

$$S_1 = \sqrt{\frac{1 - C_2}{2}}, \quad C_1 = \sqrt{\frac{1 + C_2}{2}}, \quad (81)$$

$$S_2 = \sqrt{\frac{1 - C_4}{2}}, \quad C_2 = \sqrt{\frac{1 + C_4}{2}}, \quad (82)$$

$$S_3 = S_{18}C_{15} - C_{18}S_{15}, \quad C_3 = C_{18}C_{15} + S_{18}S_{15}, \quad (83)$$

$$S_4 = S_9C_5 - C_9S_5, \quad C_4 = C_9C_5 + S_9S_5, \quad (84)$$

$$S_5 = \sqrt{\frac{1 - C_{10}}{2}}, \quad C_5 = \sqrt{\frac{1 + C_{10}}{2}}, \quad (85)$$

$$S_9 = \sqrt{\frac{1 - C_{18}}{2}}, \quad C_9 = \sqrt{\frac{1 + C_{18}}{2}}, \quad (86)$$

$$S_{10} = \sqrt{\frac{1 - R}{2}}, \quad C_{10} = \sqrt{\frac{1 + R}{2}}, \quad (87)$$

$$S_{15} = \frac{\sqrt{6} - \sqrt{2}}{4}, \quad C_{15} = \frac{\sqrt{6} + \sqrt{2}}{4}, \quad (88)$$

$$S_{18} = \frac{\sqrt{5} - 1}{4}, \quad C_{18} = \frac{\sqrt{10 + 2\sqrt{5}}}{4}, \quad (89)$$

$$S_{20} = \sqrt{1 - R^2}, \quad C_{20} = R, \quad (90)$$

$$S_{30} = \frac{1}{2}, \quad C_{30} = \frac{\sqrt{3}}{2}, \quad (91)$$

$$S_{45} = \frac{\sqrt{2}}{2}, \quad C_{45} = \frac{\sqrt{2}}{2}. \quad (92)$$

The following space was deliberately left blank.

13 Table of Integral Angles from 0° to 45°

For compactness, every table entry is expressed through already-defined radical quantities. Since every S_k and C_k appearing in the table is itself a radical expression, every entry is consequently a radical.

Table 1: Radical representations of $\sin A^\circ$ and $\cos A^\circ$ for $0^\circ \leq A \leq 45^\circ$

Part I: 0° – 22°			Part II: 23° – 45°		
Angle	$\sin(\text{angle})$	$\cos(\text{angle})$	Angle	$\sin(\text{angle})$	$\cos(\text{angle})$
0°	0	1	23°	$S_{18}C_5 + C_{18}S_5$	$C_{18}C_5 - S_{18}S_5$
1°	S_1	C_1	24°	$S_{15}C_9 + C_{15}S_9$	$C_{15}C_9 - S_{15}S_9$
2°	S_2	C_2	25°	$S_{15}C_{10} + C_{15}S_{10}$	$C_{15}C_{10} - S_{15}S_{10}$
3°	$S_{18}C_{15} - C_{18}S_{15}$	$C_{18}C_{15} + S_{18}S_{15}$	26°	$S_{30}C_4 - C_{30}S_4$	$C_{30}C_4 + S_{30}S_4$
4°	$S_9C_5 - C_9S_5$	$C_9C_5 + S_9S_5$	27°	$S_{18}C_9 + C_{18}S_9$	$C_{18}C_9 - S_{18}S_9$
5°	S_5	C_5	28°	$S_{18}C_{10} + C_{18}S_{10}$	$C_{18}C_{10} - S_{18}S_{10}$
6°	$S_3C_3 + C_3S_3$	$C_3C_3 - S_3S_3$	29°	$S_{20}C_9 + C_{20}S_9$	$C_{20}C_9 - S_{20}S_9$
7°	$S_4C_3 + C_4S_3$	$C_4C_3 - S_4S_3$	30°	S_{30}	C_{30}
8°	$S_4C_4 + C_4S_4$	$C_4C_4 - S_4S_4$	31°	$S_{30}C_1 + C_{30}S_1$	$C_{30}C_1 - S_{30}S_1$
9°	S_9	C_9	32°	$S_{30}C_2 + C_{30}S_2$	$C_{30}C_2 - S_{30}S_2$
10°	S_{10}	C_{10}	33°	$S_{18}C_{15} + C_{18}S_{15}$	$C_{18}C_{15} - S_{18}S_{15}$
11°	$S_2C_9 + C_2S_9$	$C_2C_9 - S_2S_9$	34°	$S_{30}C_4 + C_{30}S_4$	$C_{30}C_4 - S_{30}S_4$
12°	$S_3C_9 + C_3S_9$	$C_3C_9 - S_3S_9$	35°	$S_{20}C_{15} + C_{20}S_{15}$	$C_{20}C_{15} - S_{20}S_{15}$
13°	$S_9C_4 + C_9S_4$	$C_9C_4 - S_9S_4$	36°	$S_{18}C_{18} + C_{18}S_{18}$	$C_{18}^2 - S_{18}^2$
14°	$S_9C_5 + C_9S_5$	$C_9C_5 - S_9S_5$	37°	$S_{45}C_8 - C_{45}S_8$	$C_{45}C_8 + S_{45}S_8$
15°	S_{15}	C_{15}	38°	$S_{20}C_{18} + C_{20}S_{18}$	$C_{20}C_{18} - S_{20}S_{18}$
16°	$S_{15}C_1 + C_{15}S_1$	$C_{15}C_1 - S_{15}S_1$	39°	$S_{30}C_9 + C_{30}S_9$	$C_{30}C_9 - S_{30}S_9$
17°	$S_{15}C_2 + C_{15}S_2$	$C_{15}C_2 - S_{15}S_2$	40°	$S_{20}C_{20} + C_{20}S_{20}$	$C_{20}^2 - S_{20}^2$
18°	S_{18}	C_{18}	41°	$S_{45}C_4 - C_{45}S_4$	$C_{45}C_4 + S_{45}S_4$
19°	$S_{10}C_9 + C_{10}S_9$	$C_{10}C_9 - S_{10}S_9$	42°	$S_{45}C_3 - C_{45}S_3$	$C_{45}C_3 + S_{45}S_3$
20°	S_{20}	C_{20}	43°	$S_{45}C_2 - C_{45}S_2$	$C_{45}C_2 + S_{45}S_2$
21°	$S_{18}C_3 + C_{18}S_3$	$C_{18}C_3 - S_{18}S_3$	44°	$S_{45}C_1 - C_{45}S_1$	$C_{45}C_1 + S_{45}S_1$
22°	$S_{18}C_4 + C_{18}S_4$	$C_{18}C_4 - S_{18}S_4$	45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$

14 Complementary Angles

For any angle A ,

$$\sin(90^\circ - A) = \cos A, \quad (93)$$

$$\cos(90^\circ - A) = \sin A. \quad (94)$$

Consequently, the complete first quadrant follows from the lower half:

$$\boxed{\{46^\circ, \dots, 90^\circ\} = \{90^\circ - A : A = 0^\circ, \dots, 44^\circ\}.} \quad (95)$$

For example,

$$\sin 72^\circ = \cos 18^\circ, \quad (96)$$

$$\cos 72^\circ = \sin 18^\circ, \quad (97)$$

$$\sin 75^\circ = \cos 15^\circ, \quad (98)$$

$$\cos 75^\circ = \sin 15^\circ, \quad (99)$$

$$\sin 80^\circ = \cos 10^\circ, \quad (100)$$

$$\cos 80^\circ = \sin 10^\circ, \quad (101)$$

$$\sin 89^\circ = \cos 1^\circ, \quad (102)$$

$$\cos 89^\circ = \sin 1^\circ. \quad (103)$$

15 Extension to the Complete Circle

For $0^\circ \leq A \leq 360^\circ$, the quadrant relations are

$$0^\circ \leq A \leq 90^\circ : \quad \sin A = \sin A, \quad \cos A = \cos A, \quad (104)$$

$$90^\circ \leq A \leq 180^\circ : \quad \sin A = \sin(180^\circ - A), \quad \cos A = -\cos(180^\circ - A), \quad (105)$$

$$180^\circ \leq A \leq 270^\circ : \quad \sin A = -\sin(A - 180^\circ), \quad \cos A = -\cos(A - 180^\circ), \quad (106)$$

$$270^\circ \leq A \leq 360^\circ : \quad \sin A = -\sin(360^\circ - A), \quad \cos A = \cos(360^\circ - A). \quad (107)$$

Therefore every integral angle in the complete circle is generated from the first quadrant.

$$\boxed{\forall A \in \{0, 1, \dots, 360\}, \quad \sin A^\circ, \cos A^\circ \text{ are obtained from the radical anchor set.}} \quad (108)$$

16 Proof of Radical Closure

Let $\mathbb{K}$ be the field generated by the radical quantities appearing in the construction:

$$\mathbb{K} = \mathbb{Q} \left(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{10 + 2\sqrt{5}}, R \right), \quad (109)$$

where R is itself generated by the conjugate cubic radicals

$$R = \sqrt[3]{\frac{1 + i\sqrt{3}}{16}} + \sqrt[3]{\frac{1 - i\sqrt{3}}{16}}. \quad (110)$$

The construction then repeatedly applies:

1. field addition;
2. field subtraction;
3. field multiplication;
4. field division by nonzero elements;
5. extraction of square roots;
6. extraction of cube roots.

Each operation preserves radical constructibility. Hence every generated quantity belongs to a radical extension of $\mathbb{Q}$.

The trigonometric addition law also preserves the Pythagorean constraint. Indeed,

$$(\sin \alpha \cos \beta + \cos \alpha \sin \beta)^2 + (\cos \alpha \cos \beta - \sin \alpha \sin \beta)^2 \quad (111)$$

$$= (\sin^2 \alpha + \cos^2 \alpha)(\sin^2 \beta + \cos^2 \beta) = 1. \quad (112)$$

Thus the addition law is algebraically compatible with the unit-circle constraint.

17 Geometric Interpretation

The unit-circle representation is

$$\mathbf{u}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}. \quad (113)$$

The addition law may be represented as a rotation:

$$\mathbf{u}(\alpha + \beta) = \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix} \mathbf{u}(\alpha). \quad (114)$$

A schematic of the constructive hierarchy is shown below.

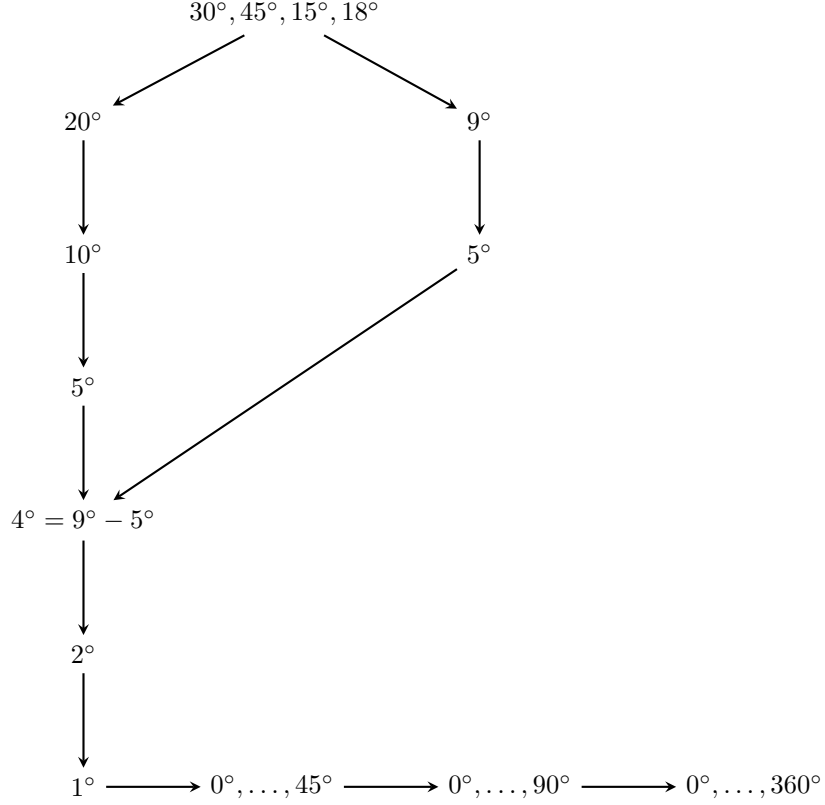


Figure 1: Schematic of the constructive hierarchy.

The apparent duplicate 5° nodes emphasize that 5° is obtained from the 10° branch and then participates in the subtraction construction for 4° .

18 Algorithmic Formulation

The mathematical construction can be implemented as a deterministic symbolic algorithm.

$$\begin{aligned}
& \text{CONSTRUCTRADICALS}() \\
& R \leftarrow \sqrt[3]{(1 + i\sqrt{3})/16} + \sqrt[3]{(1 - i\sqrt{3})/16} \\
& C_{20} \leftarrow R \\
& S_{20} \leftarrow \sqrt{1 - R^2} \\
& S_{10} \leftarrow \sqrt{(1 - R)/2} \\
& C_{10} \leftarrow \sqrt{(1 + R)/2} \\
& S_5 \leftarrow \sqrt{(1 - C_{10})/2} \\
& C_5 \leftarrow \sqrt{(1 + C_{10})/2} \\
& C_{18} \leftarrow \sqrt{10 + 2\sqrt{5}}/4 \\
& S_{18} \leftarrow (\sqrt{5} - 1)/4 \\
& S_9 \leftarrow \sqrt{(1 - C_{18})/2} \\
& C_9 \leftarrow \sqrt{(1 + C_{18})/2} \\
& S_4 \leftarrow S_9 C_5 - C_9 S_5 \\
& C_4 \leftarrow C_9 C_5 + S_9 S_5 \\
& S_2 \leftarrow \sqrt{(1 - C_4)/2} \\
& C_2 \leftarrow \sqrt{(1 + C_4)/2} \\
& S_1 \leftarrow \sqrt{(1 - C_2)/2} \\
& C_1 \leftarrow \sqrt{(1 + C_2)/2} \\
& \text{return all generated quantities.}
\end{aligned} \tag{115}$$

The remaining angles are produced by addition, subtraction, complementary mapping, and quadrant mapping.

19 Verification and Numerical Stability

A numerical implementation should verify the invariant

$$\varepsilon_A = |\sin^2 A + \cos^2 A - 1|. \quad (116)$$

A second verification condition is

$$\delta_A = |\sin A - \sin(A)|, \quad (117)$$

where the second representation is independently evaluated numerically.

For practical computation, high-precision arithmetic should be used because nested radicals and cancellation can amplify floating-point errors.

The computational validation protocol is:

$$\begin{aligned} &\text{VERIFY}(A) \\ &\quad s \leftarrow \text{radical expression for } \sin A \\ &\quad c \leftarrow \text{radical expression for } \cos A \\ &\quad e_1 \leftarrow |s^2 + c^2 - 1| \\ &\quad e_2 \leftarrow |s - \sin_{\text{numeric}}(A)| \\ &\quad e_3 \leftarrow |c - \cos_{\text{numeric}}(A)| \\ &\quad \textbf{accept if } e_1, e_2, e_3 < \varepsilon. \end{aligned} \quad (118)$$

This separates symbolic correctness from numerical implementation error.

20 Statistics and Data Science Interpretation

The construction may be viewed as a deterministic data-generation system. The “data” are the initial exact anchors, the “features” are the algebraic generators, and the “transformations” are addition, subtraction, and half-angle maps.

A symbolic dependency graph has the form

$$\text{anchors} \longrightarrow \text{algebraic generators} \longrightarrow \text{derived angles} \longrightarrow \text{validation}. \quad (119)$$

This resembles a data lineage system in which every derived observation has an auditable provenance.

Unlike statistical inference, however, the present problem contains no sampling uncertainty. The quantities are deterministic. Thus the appropriate statistical concepts are error propagation, numerical conditioning, reproducibility, and validation rather than confidence intervals or hypothesis tests.

21 Computer Science and AI Interpretation

The radical construction is naturally represented as a directed acyclic graph of symbolic dependencies. Each node contains an exact algebraic expression, while each edge represents an allowed transformation.

The state of the symbolic system can be represented as

$$\mathcal{S} = \{S_A, C_A : A \in \mathcal{A}\}. \quad (120)$$

A transformation function

$$T(\alpha, \beta) = (S_\alpha C_\beta + C_\alpha S_\beta, C_\alpha C_\beta - S_\alpha S_\beta) \quad (121)$$

implements angle addition.

From an AI/ML perspective, this is a symbolic rather than statistical learning problem. There is no need to estimate unknown parameters from examples. The system performs exact symbolic reasoning under algebraic constraints.

A useful distinction is therefore:

$$\boxed{\text{machine learning} \neq \text{symbolic algebra}} \quad (122)$$

although modern AI systems can assist with both.

22 Physics Interpretation

The vector

$$\mathbf{u}(\theta) = (\cos \theta, \sin \theta) \quad (123)$$

is a unit vector. Rotation preserves its norm:

$$\|\mathbf{u}(\theta)\|^2 = \cos^2 \theta + \sin^2 \theta = 1. \quad (124)$$

Thus the Pythagorean identity is an invariant under planar rotation.

The same mathematical structure occurs in harmonic oscillation:

$$x(t) = A \cos(\omega t + \phi), \quad (125)$$

$$v(t) = -A\omega \sin(\omega t + \phi). \quad (126)$$

The present radical construction therefore illustrates a general physical principle: complicated states may be generated by composition of elementary transformations while preserving an invariant.

23 Chemistry and Molecular Geometry

Many molecular-geometry calculations involve angles whose trigonometric components enter coordinate systems, dot products, bond-angle calculations, and rotational transformations.

For vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta. \quad (127)$$

Thus exact radical trigonometric values can, when the geometry permits, provide exact symbolic coordinates and exact dot products.

This is especially useful conceptually for highly symmetric molecular structures, although practical computational chemistry generally uses floating-point numerical methods.

24 Biology and Geometry of Periodic Systems

Biological systems contain periodic or quasi-periodic structures at many scales, including rotational symmetry, cyclic molecular structures, and oscillatory biological signals.

The mathematical lesson is not that biological phenomena are generated by these particular radicals, but that periodic systems can often be represented through a compact set of generators and transformation rules.

25 Psychology and Psychiatry

Periodic mathematical representations can also appear in computational models of rhythmic behaviour, oscillatory signals, and cyclic processes.

However, no clinical or psychological inference follows from the present trigonometric construction. The relevant methodological analogy is instead the distinction between:

$$\text{state representation} \quad \text{and} \quad \text{empirical inference}. \quad (128)$$

The present paper concerns exact state representation.

26 Economics and Finance

Financial mathematics frequently uses trigonometric functions in Fourier methods, harmonic decompositions, seasonal models, and characteristic functions.

A generic periodic economic signal may be represented as

$$X(t) = a_0 + \sum_{k=1}^N [a_k \cos(k\omega t) + b_k \sin(k\omega t)]. \quad (129)$$

Exact trigonometric values can therefore enter symbolic calculations of periodic components.

In finance, Fourier methods appear in characteristic-function approaches to option pricing. The present construction does not constitute an asset pricing model, but it illustrates the broader computational principle of reducing complicated calculations to reusable algebraic primitives.

27 Welfare Economics

A welfare function may be viewed abstractly as a mapping

$$W = W(x_1, \dots, x_n). \quad (130)$$

The trigonometric construction illustrates a different but related principle: a complex aggregate can be constructed from primitive components while maintaining explicit provenance.

This is useful conceptually when studying decomposability, aggregation, and transformation invariance.

28 Political Science, Geopolitics and International Relations

A system of states can be represented abstractly as a network

$$G = (V, E), \quad (131)$$

where vertices represent states and edges represent relations.

The radical construction has an analogous dependency-network interpretation:

$$G_{\text{radical}} = (V_{\text{angles}}, E_{\text{transformations}}). \quad (132)$$

The analogy is structural only. Trigonometric identities do not imply political outcomes.

Nevertheless, the distinction between primitive nodes, transformation rules, and derived states is useful in formal systems analysis.

29 Diplomacy and International Systems

Addition and subtraction formulae provide a deterministic composition law:

$$T(\alpha, \beta) = \alpha + \beta. \quad (133)$$

This may be regarded abstractly as a composition operator in a state space. A diplomatic or institutional model can similarly distinguish:

1. primitive states;
2. transformation rules;
3. derived states;
4. invariants;
5. validation conditions.

Again, this is a systems-engineering analogy rather than a substantive theory of diplomacy.

30 Warfare Economics and Warfare

In military systems engineering, capability is often decomposed into components which interact through explicit transformation rules.

A generic state-transition representation is

$$x_{t+1} = F(x_t, u_t). \quad (134)$$

The present trigonometric problem has an exact analogous structure:

$$x_{\text{new}} = F(x_{\text{old}}, \theta). \quad (135)$$

The key methodological lesson is that compositional systems should preserve invariants and expose dependencies. In the present case the invariant is the unit-circle norm. In an engineering system the invariant could instead be conservation, capacity, safety, or resource balance.

No operational military recommendation follows from the trigonometric construction.

31 Philosophical Interpretation

The construction illustrates a classical distinction between:

$$\text{discovery} \quad \text{and} \quad \text{construction.} \quad (136)$$

The numerical value of $\sin 1^\circ$ may be discovered by numerical approximation, but the present method constructs it from explicit algebraic operations.

This reflects the constructive mathematical principle that an existence statement is stronger when accompanied by an explicit procedure producing the object.

32 Religious and Classical Mathematical Perspective

The construction also illustrates a recurring mathematical theme found across ancient and classical traditions: complicated quantities can emerge from simple relations, symmetries, and invariant structures.

The present argument makes no theological claim. Its relevant philosophical content is the distinction between an observed quantity and a reproducible construction.

33 Systems Engineering Interpretation

The complete system can be summarized as

$$\boxed{\text{Radical Anchors} \rightarrow \text{Cubic} \rightarrow \text{Quadratics} \rightarrow \begin{matrix} \text{Addition/} \\ \text{Subtraction} \end{matrix} \rightarrow \text{Complementation} \rightarrow \begin{matrix} \text{Quadrant} \\ \text{Extension} \end{matrix}} \quad (137)$$

Each layer has a clear responsibility:

Table 2: Layers and their functions.

Layer	Function
Anchor layer	supplies elementary exact radicals
Cubic layer	constructs $\cos 20^\circ$
Quadratic layer	performs half-angle reductions
Composition layer	constructs sums and differences
Symmetry layer	constructs complementary angles
Quadrant layer	determines signs over the unit circle
Verification layer	checks algebraic and numerical invariants

34 Complete-Circle Theorem

Theorem. For every integral A satisfying

$$0 \leq A \leq 360, \quad (138)$$

the quantities

$$\sin A^\circ \quad \text{and} \quad \cos A^\circ \quad (139)$$

are obtainable by a finite sequence of rational operations, square roots, cube roots, and the conjugate complex radicals appearing in the Cardano representation of $\cos 20^\circ$.

Proof. The values at 30° , 45° , 15° , and 18° are radical quantities. The cubic

$$8x^3 - 6x - 1 = 0 \quad (140)$$

constructs $\cos 20^\circ$ by Cardano's formula. The half-angle equations then construct 10° , 5° , 4° , 2° , and 1° . The subtraction formula constructs 3° from 18° and 15° .

Every integer angle from 0° through 45° can then be expressed as a finite sum or difference of the constructed angles. The addition and subtraction identities consequently produce its sine and cosine.

For angles from 46° through 90° , the complementary identities

$$\sin(90^\circ - A) = \cos A, \quad (141)$$

$$\cos(90^\circ - A) = \sin A \quad (142)$$

reduce the problem to the preceding interval.

Finally, the quadrant identities reduce every angle from 91° through 360° to an angle in the first quadrant, with the appropriate signs. All operations are finite radical operations. Therefore the proposition follows. $\square$

35 Complexity and Reproducibility

The construction is not an algorithm for numerical approximation in the usual sense. Its objective is symbolic exactness.

If each radical expression is represented as a node in an expression DAG, common subexpressions such as R , S_9 , C_9 , S_5 , and C_5 need not be recomputed.

The computational architecture is therefore

$$\text{symbolic DAG} + \text{memoization} + \text{invariant checking}. \quad (143)$$

This is preferable to independently expanding every radical expression, because unrestricted expansion can cause severe expression swell.

36 Error Analysis

Although the mathematical expressions are exact, numerical evaluation can be unstable if performed with insufficient precision.

For a differentiable function f ,

$$\delta f \approx f'(x)\delta x. \quad (144)$$

Nested square roots and subtraction can magnify small numerical errors. Therefore a practical implementation should use guard digits and verify

$$\sin^2 A + \cos^2 A = 1 \quad (145)$$

after evaluation.

For exact symbolic work, the preferred representation is the unreduced expression tree rather than a floating-point approximation.

37 Conclusion

The central result is constructive.

A complete integral-degree trigonometric table does not require 361 independent polynomial solutions. A small set of algebraic generators, combined with the group structure of rotations, is sufficient.

The principal chain is

$30^\circ, 45^\circ, 15^\circ, 18^\circ$	20°	9°
	$\downarrow$	$\downarrow$
	10°	
	$\downarrow$	
	5°	
	$\searrow$	
	4°	
	$\downarrow$	
	2°	
	$\downarrow$	
	1°	

(146)

followed by

$$[0^\circ, 45^\circ] \longrightarrow [0^\circ, 90^\circ] \longrightarrow [0^\circ, 360^\circ]. \quad (147)$$

The essential mathematical ingredients are:

1. the Pythagorean identity;
2. half-angle formulae;
3. triple-angle cubics;
4. Cardano's radical solution;
5. addition and subtraction formulae;
6. complementary-angle identities;
7. quadrant symmetries;
8. exact algebraic closure.

The deeper computational lesson is that an apparently high-degree exact evaluation problem may possess a much smaller compositional representation. The appropriate object is therefore not merely a list of numerical values, but an auditable algebraic dependency system.

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Glossary

Anchor angle

An angle whose sine and cosine are already available in exact radical form and which is used to construct further angles.

Casus irreducibilis

The situation in which a cubic has three real roots but Cardano's formula passes through complex radicals.

Complementary angle

For an angle A , the angle $90^\circ - A$.

Conjugate radicals

Complex radical expressions occurring in conjugate pairs, whose sum may be real.

Expression DAG

A directed acyclic graph representation of a symbolic expression in which common subexpressions can be shared.

Field extension

An enlargement of a base field obtained by adjoining algebraic quantities.

Half-angle formula

An identity expressing trigonometric functions of $\theta/2$ through trigonometric functions of θ .

Invariant

A quantity preserved by a transformation. Here the principal invariant is $\sin^2 \theta + \cos^2 \theta = 1$.

Quadrant symmetry

The sign and reflection structure of sine and cosine over the four quadrants of the unit circle.

Radical expression

An algebraic expression obtained using arithmetic operations and extraction of roots.

Triple-angle formula

An identity expressing $\sin 3\theta$ or $\cos 3\theta$ as a polynomial in $\sin \theta$ or $\cos \theta$.

Unit circle

The circle of radius one centered at the origin, parametrized by $(\cos \theta, \sin \theta)$.

The End